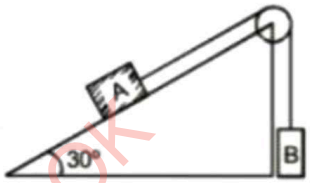
Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two blocks are connected over a massless pulley as shown in fig. The mass of block A is 10 kg and the coefficient of kinetic friction is 0.2. Block A slides down the incline at constant speed. The mass of block B in kg is:



A. 3.5

B. 3.3

C. 3

D. 2.5

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When a body moves in a circular path, no work is done by the force since

A. force and displacement are perpendicular to each other

B. the force is always away from the centre

C. there is no displacement

D. there is no net force

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The potential energy for a force field $\vec{F}$ is given by $U(x, y) = \cos(x + y)$. The force acting on a particle at position given by coordinates $(0, \pi/4)$ is $-$

A. $-\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$

B. $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$

C. $\left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)$

D. $\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A body of mass 6 kg moves in a straight line according to the equation, $x = t^3 - 75t$, where x denotes the distance in metre and t the time in second. The force on the body at $t = 4$ s is

A. 64 N

B. 72 N

C. 144 N

D. 36 N

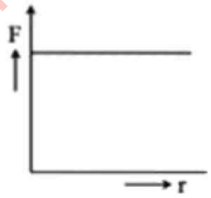
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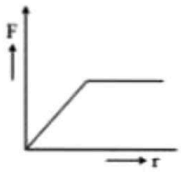
Question No. 5

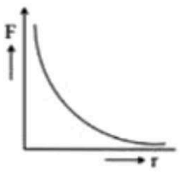
Only One Option Correct Type

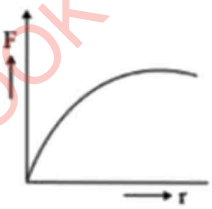
Each question has multiple options out of which ONLY ONE is correct.

If a particle moving in a circular path maintains a constant speed of $20\ \text{ms}^{-1}$, then which of the following correctly describes relation between force (F) and radius (r) ?

A. 

B. 

C. 

D. 

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Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A body of mass $5\ \text{kg}$ is thrown vertically up with a kinetic energy of $490\ \text{J}$. The height at which the kinetic energy of the body becomes half of the original value is
(acceleration due to gravity $= 9.8\ \text{ms}^{-2}$)

A. $5\ \text{m}$

B. $2.5\ \text{m}$

C. $10\ \text{m}$

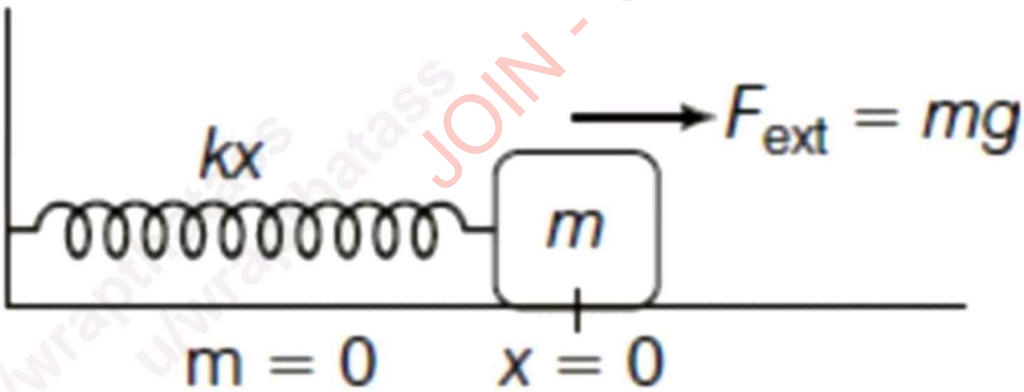
D. $12.5\ \text{m}$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A force starts acting on a spring block system. At a stretch when $x = \frac{mg}{k}$, workdone by external force and spring force are related as



A. $W_{F_{ext}} + W_{F_s} = 0$

B. $W_{F_{ext}} = \frac{1}{2}\,kx^2$

C. $W_{F_s} + W_{F_{ext}} > 0$

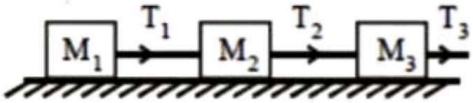
D. None of these

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three blocks of masses m_1, m_2 and m_3 are connected by massless strings, as shown, on a frictionless table. They are pulled with a force $T_3 = 40$ N. If $m_1 = 10$ kg, $m_2 = 6$ kg and $m_3 = 4$ kg, the tension T_2 will be



A. 20 N

B. 40 N

C. 10 N

D. 32 N

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The upper half of an inclined plane with inclination ϕ is perfectlysmooth while the lower half is rough. A body starting from rest at the top will again come to rest at the bottom if the coefficient of friction for the lower half is given by

A. $2\cos\phi$

B. $2\sin\phi$

C. $\tan\phi$

D. $2\tan\phi$

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u/wrapthatass

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following must be known in order to determine the power output of an automobile?

A. Final velocity and height

B. Mass and amount of work performed

C. Force exerted and distance of motion

D. Work performed and elapsed time of work

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A body of mass m is moving in a straight line with momentum p. Starting at time $t = 0$, a force $F = at$ acts in the same direction on the moving particle during time interval of T. So that its momentum changes from p to $2p$. The value of T is

A. $\sqrt{\frac{2p}{a}}$

B. $\sqrt{\frac{p}{a}}$

C. $2\sqrt{\frac{2p}{a}}$

D. $\frac{2p}{a}$

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A boy pushes a toy box 2.0 m along the floor by means of a force of 10 N directed downward at an angle of 60° to the horizontal. The work done by the boy is

A. 6 J

B. 8 J

C. 10 J

D. 12 J

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A plate of mass (M) is placed on a horizontal frictionless surface and a body of mass (m) is placed on this plate. The coefficient of dynamic friction between this body and the plate is μ. If a force $3\mu\mathrm{mg}$ is applied to the body of mass (m) along the horizontal, the acceleration of the plate will be

A. $\frac{\mu \mathrm{m}}{\mathrm{M}}\ \mathrm{g}$

B. $\frac{\mu \mathrm{mg}}{\mathrm{M}+\mathrm{m}}$

C. $\frac{3\mu \mathrm{mg}}{\mathrm{M}}$

D. $\frac{2\mu \mathrm{mg}}{\mathrm{M}+\mathrm{m}}$

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An elastic string of unstretched length L and force constant k is stretched by a small length x. It is further stretched by another small length y. The work done in the second stretching is

A. $1/2\mathrm{Ky}^2$

B. $1/2\mathrm{Ky}(2\mathrm{x}+\mathrm{y})$

C. $1/2\ \mathrm{K}(\mathrm{x}^2+\mathrm{y}^2)$

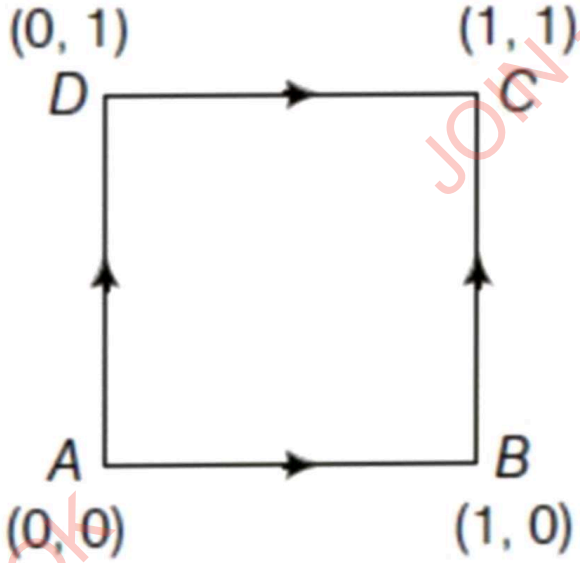
D. $1/2\mathrm{k}(\mathrm{x}+\mathrm{y})^2$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle is moving in a force field given by $\mathbf{F}=y^2\hat{\mathbf{i}}-x^2\hat{\mathbf{j}}$. Starting from A, the particle has to reach C either along ABC or ADC. Let the workdone along the two paths be W_1 and W_2, respectively. Then, (W_1, W_2) are



A. $(-1,1)$

B. $(-1,0)$

C. $(1,1)$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

At any instant the velocity of a particle of mass 500 g is $\left(2t\hat{i}+3t^2\hat{j}\right)\text{ms}^{-1}$. If the force acting on the particle at $t=1s$ is $(\hat{i}+x\hat{j})N$. Then the value of x will be:

A. 3

B. 4

C. 6

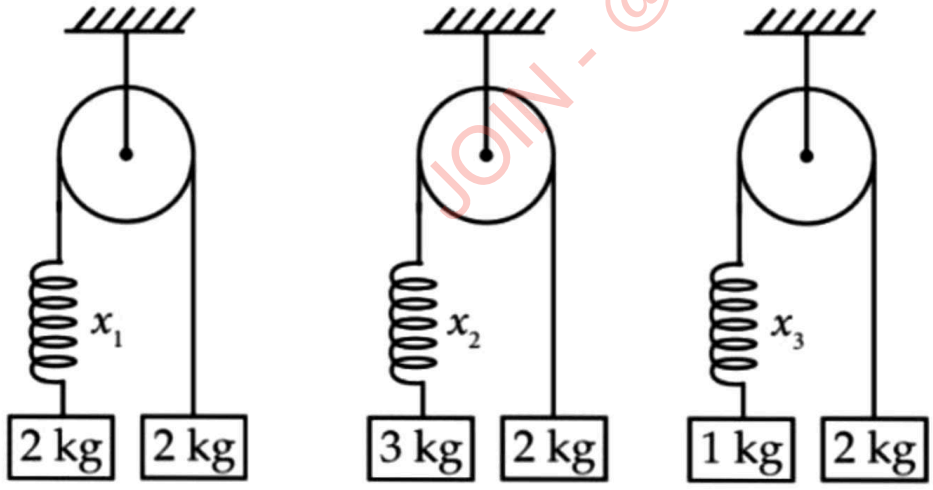
D. 2

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Same spring is attached with 2 kg, 3 kg and 1 kg blocks in three different cases as shown in figure. If x_1, x_2, x_3 be the extensions in the spring in the three cases, then



A. $x_1=0, x_3>x_2$

B. $x_1>x_2>x_3$

C. $x_3>x_2>x_1$

D. $x_2>x_1>x_3$

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The minimum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is

A. 60

B. 30

C. 15

D. 25

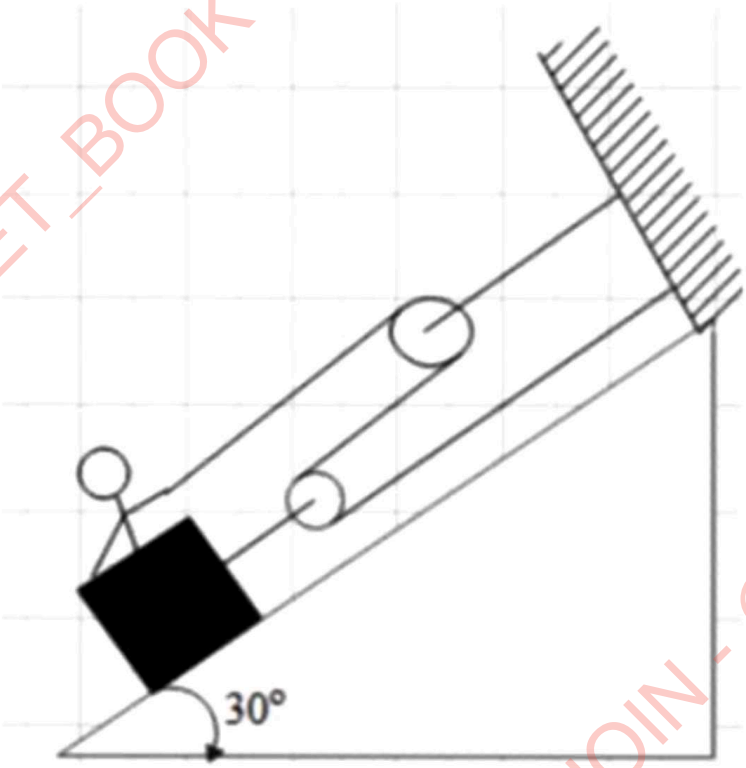
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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A man pulls himself up the 30° incline by the method shown in figure. If the combined mass of the man and cart is $100\ \text{kg}$, determine the acceleration of the cart if the man exerts a pull of $250\ \text{N}$ on the rope. Neglect all friction and the mass of the rope, pulleys, and wheels.



A. $4.5\ \text{m s}^{-2}$

B. $2.5\ \text{ms}^{-2}$

C. $3.5\ \text{m s}^{-2}$

D. $1.5\ \text{m s}^{-2}$

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A smooth chain of length $2\ \text{m}$ is kept on a table such that its length of $60\ \text{cm}$ hangs frecly from the edge of the table. The total mass of the chain is $4\ \text{kg}$. The work done in pulling the entire chain on the table is (Take, $g = 10\ \text{m/s}^2$)

A. $6.3\ \text{J}$

B. $3.6\ \text{J}$

C. $2.0\ \text{J}$

D. $12.9\ \text{J}$

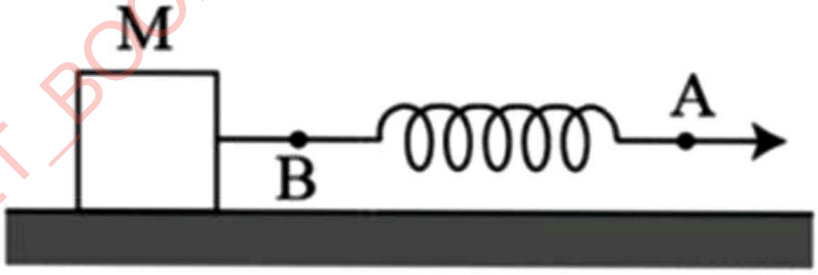
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A block of mass M is placed on a smooth horizontal surface and it is pulled by a light spring as shown in the diagram. If the ends A and B of the spring are moving with $4\ \text{m s}^{-1}$ and $2\ \text{m s}^{-1}$ respectively and the rate at which spring energy is increasing is $20\ \text{J s}^{-1}$. then what is the value of spring force (in N)?



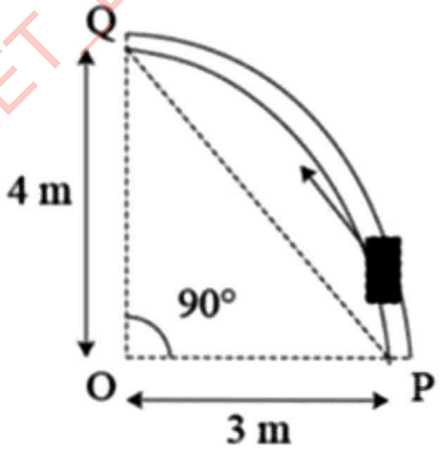
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Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Consider an elliptically shaped rail PQ in the vertical plane with $\mathrm{OP} = 3\ \mathrm{m}$ and $\mathrm{OQ} = 4\ \mathrm{m}$. A block of mass $1\ \mathrm{kg}$ is pulled along the rail from P to Q with a force of $18\ \mathrm{N}$, which is always parallel to line PQ (see) the figure given). Assuming no frictional losses, the kinetic energy of the block when it reaches Q is $(\mathrm{n} \times 10)$ joules. The value of n is (take acceleration due to gravity $= 10\ \mathrm{m\ s^{-2}}$)



Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

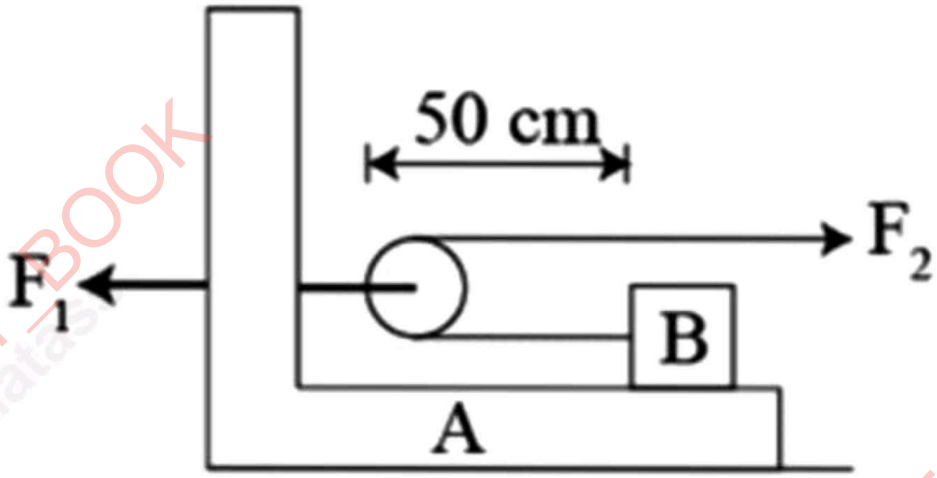
A small object of mass of $100\ \mathrm{g}$ moves in a circular path. At a given instant velocity of the object is $10\hat{i}\ \mathrm{m/s}$ and acceleration is $\left(20\hat{i} + 10\hat{j}\right)\ \mathrm{m/s^2}$. At this instant of time, rate of change of kinetic energy (in $\mathrm{kg\ m^2\ s^{-3}}$) of the object is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A 1 kg block B rests as shown on a bracket A of the same mass. Constant forces $F_1 = 20$ N and $F_2 = 8$ N start to act at a time $t = 0$ when the distance of block B from the pulley is 50 cm. Time (in s) when block B reaches the pulley is t. Find the value of $10t$.

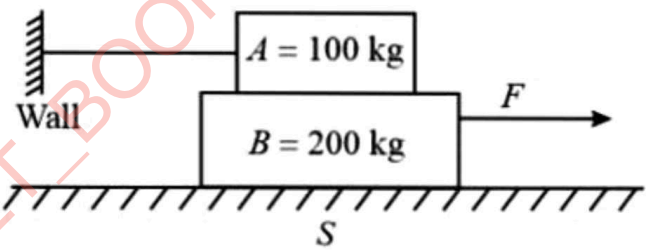


Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

A block A of mass 100 kg rests on another block B of mass 200 kg and is tied to a wall as shown in the figure. The coefficient of friction between A and B is 0.2 and that between B and ground is 0.3. The minimum force required to move the block B is $(g = 10 \text{ m s}^{-2})$



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Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The increasing order of the strength of hydrogen bond in the following mentioned linkages is:

(i) $O - H \cdots S$

(ii) $S - H \cdots O$

(iii) $F - H \cdots F^-$

(iv) $F - H \cdots O$

A. $(i) < (ii) < (iii) < (iv)$

B. $(i) < (ii) < (iv) < (iii)$

C. $(ii) < (i) < (iii) < (iv)$

D. None of these

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When two compounds ACl_3 and DCl_3 of two elements A and D are mixed together a compound $ADCl_6$ is formed. Structural analysis showed that $ADCl_6$ is an ionic compound. Given that DCl_3 is trigonal planar and ACl_3 is trigonal pyramidal, predict the shape of the anion DCl_4^-.

A. See-saw

B. Perfect tetrahedral

C. Square planar

D. None of these

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Arrange the given species F, F^-, O and O^{-2} in the order of their radii

A. $O^{2-} > F^- > O > F$

B. $O^{2-} > F^- > F > O$

C. $F^- > O^{2-} > F > O$

D. $O^{2-} > O > F^- > F$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the correct match for the column I, from the column II and choose the correct option:

Column - I	**Column - II**
(a) C_2H_2	(i) sp^3d^2 hybridisation
(b) SF_6	(ii) sp^3d^3 hybridisation
(c) SO_2	(iii) sp hybridisation
(d) IF_7	(iv) sp^2 hybridisation

A. (a) $\longrightarrow$ (i), (b) $\longrightarrow$ (iii), (c) $\longrightarrow$ (ii), (d) $\longrightarrow$ (iv)

B. (a) $\longrightarrow$ (iii), (b) $\longrightarrow$ (i), (c) $\longrightarrow$ (iv), (d) $\longrightarrow$ (ii)

C. (a) $\longrightarrow$ (ii), (b) $\longrightarrow$ (iii), (c) $\longrightarrow$ (i), (d) $\longrightarrow$ (iv)

D. (a) $\longrightarrow$ (iv), (b) $\longrightarrow$ (i), (c) $\longrightarrow$ (iii), (d) $\longrightarrow$ (ii)

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When N_2 is converted into N_2^+ and O_2 is converted into O_2^+, then the bond strength, respectively.

A. increases, increases

B. decreases, decreases

C. increases, decreases

D. decreases, increases

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What was the main drawback of Mendeleev's periodic table?

A. Many elements have several isotopes.

B. Noble gases do not form compounds.

C. Some groups stand divided into two sub groups A and B.

D. Atomic weights of elements are not always whole numbers.

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statement is correct regarding following process?

(i) $Cl \xrightarrow{EA} Cl^-$

(ii) $Cl^- \xrightarrow{IE} Cl$

(iii) $Cl \xrightarrow{IE} Cl^+$

(iv) $Cl \xrightarrow{IE} Cl^{2+}$

A. $|IE$ of Process (iv) $| = |IE$ of Process (iii) $|$

B. $|IE$ of Process (iii) $| = |IE$ of Process (ii) $|$

C. $|IE$ of Process (ii) $| = |EA$ of Process (i) $|$

D. $|IE$ of Process (iv) $| = |EA$ of Process (i) $|$

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Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The total number of covalent bonds formed in the Lewis structures of given species respectively are:

(a) CO_3^{2-}

(b) CCl_4

(c) NF_3

(d) HNO_3

A. $4,\ 4,\ 3,\ 5.$

B. $4,\ 3,\ 5,\ 4.$

C. $3,\ 5,\ 4,\ 3.$

D. $5,\ 3,\ 4,\ 5.$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following is incorrect?

A. IE_1 of $Li < IE_1$ of Be

B. IE_1 of $Be < IE_1$ of B

C. IE_1 of $Li > IE_1$ of Na

D. IE_1 of $He > IE_1$ of Ne

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Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Among the statement (I – IV), the correct ones are:

(I) Be has smaller atomic radius compared to Mg.

(II) Be has higher ionization enthalpy than AI.

(III) Charge/radius ratio of Be is greater than that of Al.

(IV) Both Be and Al form mainly covalent compounds

A. (I), (II) and (IV)

B. (I), (III) and (IV)

C. (II), (III) and (IV)

D. (I), (II) and (III)

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What will be the magnitude of the charge on the oxygen atom in water molecule if dipole moment of water is $1.85\ \mathrm{D}$ and bond length of $\mathrm{O}-\mathrm{H}$ is $0.94\ \text{Å}$?

A. $3.23 \times 10^{-10}\mathrm{esu}$

B. $0.323 \times 10^{-10}\mathrm{esu}$

C. $2.32 \times 10^{-10}\mathrm{esu}$

D. $3.32 \times 10^{-7}\mathrm{esu}$

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Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The boiling point of alkyl halide are higher than those of corresponding alkanes because of

A. dipole-dipole interaction

B. dipole-induced dipole interaction

C. H-bonding

D. None of the above

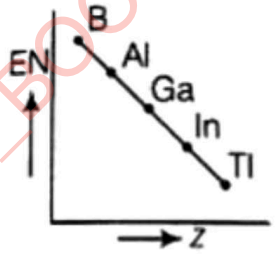
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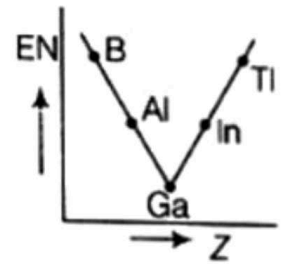
Question No. 13

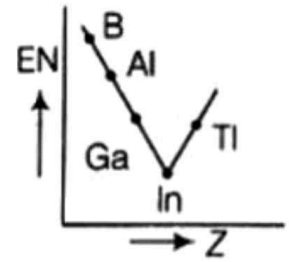
Only One Option Correct Type

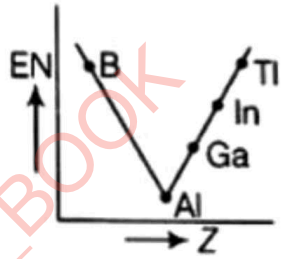
Each question has multiple options out of which ONLY ONE is correct.

Which one of the following correctly represents the variation of electronegativity (EN) with atomic number (Z) of group 13 elements?

A. 

B. 

C. 

D. 

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Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The Cl- C - Cl angle in $1,1,2,2$ -tetrachloroethene and tetrachloromethane respectively will be about

A. $109 \cdot 5^{\circ}$ and 90°

B. 120° and $109 \cdot 5^{\circ}$

C. 90° and $109 \cdot 5^{\circ}$

D. $109 \cdot 5^{\circ}$ and 120°

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the following Assertion and Reason, the correct option is:
Assertion: For hydrogenation reactions, the catalytic activity increases from Group 5 to Group 11 metals with maximum activity shown by Group $7-9$ elements.
Reason: The reactants are most strongly adsorbed on group $7-9$ elements.

A. The assertion is true, but the reason is false

B. Both assertion and reason are false

C. Both assertion and reason are true and the reason is the correct explanation for the assertion

D. Both assertion and reason are true but the reason is not the correct explanation for the assertion

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

(X), (Y), (Z) are elements of the third period. Oxide of (X) is ionic, that of (Y) is amphoteric and of (Z) a giant molecule. (X), (Y) and (Z) will have atomic numbers in the order

A. (X) < (Y) < (Z)

B. (Z) < (Y) < (X)

C. (X) < (Z) < (Y)

D. (Y) < (X) < (Z)

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The bond order in NO is 2.5 while that in NO^+ is 3. Which statement is true for these two species:

A. Bond length in NO^+ is greater than that in NO

B. Bond length in NO is greater than that in NO^+

C. Bond length is NO^+ is equal to that in NO

D. Bond length is unpredictable

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Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the columns I, II and III and mark the appropriate choice.

Column I	Column II	Column III
(a) Bromine	(i) Noble metal	(p) Amalgam
(b) Gold	(ii) Crystalline non-metal	(q) $4s^2 4p^5$
(c) Mercury	(iii) Liquid non-metal	(r) Transition metal
(d) Iodine	(iv) Liquid metal	(s) Violet

A. (a) -(iii, q), (b) -(i, r), (c) -(iv, p), (d) -(ii, s)

B. (a) -(ii, p), (b) -(i, s), (c) -(iii, q), (d) -(iv, r)

C. (a) -(i, s), (b) -(ii, p), (c) -(iv, r), (d) -(iii, q)

D. (a) -(iv, r), (b) -(iii, q), (c) -(ii, s), (d) -(i, p)

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the properties mentioned, the correct trend for the different species is in

A. Strength as Lewis acid $-BCl_3 > AlCl_3 > GaCl_3$

B. Inert pair effect $-Al > Ga > In$

C. Oxidising property $-Al^{3+} > In^{3+} > Ti^{3+}$

D. First ionisaiton enthalpy $-B > Tl > Al$

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Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Choose the one molecule or ion from each set having smallest bond angle.
(i) NH_3, PH_3 or AsH_3
(ii) O_3^+, O_3
(iii) NO_2^- or O_3
(iv) $X - S - X$ angle in $SOCl_2$ or SOF_2

A. $NH_3, O_3^+ O_3, SOCl_2$

B. $AsH_3, O_3, NO_2^-, SOF_2$

C. PH_3, O_3^+, NO_2^-

D. AsH_3, O_3^+, O_3, SOF_2

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

How many of the following have greater Z_{eff} than Silicon atom:

(i) Na (ii) Mg (iii) Al (iv) P (v) Cl (vi) S (vii) N (viii) O (ix) F

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

In PO_4^{3-}, the formal charge on each oxygen atom and the $P - O$ bond order respectively are A and B respectively, find value of $4A + 4B$.

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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

How many of these molecules are diamagnetic and have bond order more than 2

$O_2^{2+}, CO, \bar{C}N, NO, NO^+, N_2, O_2^+, N_2^{2-}, O_2^{2-}$

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Element with electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^3$ belongs to the following group of the periodic table

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

Find the number of molecule having two lone pairs on central atom.

(i) I_3^+ (ii) XeF_2 (iii) XeF_4 (iv) H_2O (v) NH_2^- (vi) H_2S (vii) H_2SO_4 (viii) NF_3

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a sequence of 21 terms, the first 11 terms are in AP with common difference 2 and the last 11 terms are in GP with common ratio 2. If the middle term of AP be equal to the middle term of the GP, then the middle term of the entire sequence is

A. $-\frac{10}{31}$

B. $\frac{10}{31}$

C. $\frac{32}{31}$

D. $-\frac{31}{32}$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $0 < A < B < \pi$, $\sin A + \sin B = \sqrt{\frac{3}{2}}$ and $\cos A + \cos B = \frac{1}{\sqrt{2}}$, then $A =$

A. 15°

B. 30°

C. 45°

D. $22\frac{1}{2}^\circ$

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Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $a, b, c \in R^+$ such that $a + b + c = 27$, then the maximum value of $a^2b^3c^4$ is equal to

A. $2^8 \cdot 3^{10}$

B. $2^9 \cdot 3^{11}$

C. $2^{10} \cdot 3^{12}$

D. $2^{11} \cdot 3^{13}$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When 2^{301} is divided by 5, the least positive remainder is :

A. 4

B. 8

C. 2

D. 6

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Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of the sum in the 50th bracket of
$(1) + (2+3) + (4+5+6) + (7+8+9+10) + \ldots$ is

A. 62525

B. 65225

C. 56255

D. 555625

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider two positive numbers a and b. If arithmetic mean of a and b exceeds their geometric mean by $\frac{3}{2}$ and geometric mean of a and b exceeds their harmonic mean by $\frac{6}{5}$, then the absolute value of $\left(a^2 - b^2\right)$ is equal to :

A. 153

B. 135

C. 154

D. 136

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Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The greatest integer less than $\left(\sqrt{2}+1\right)^6$ is equal to

A. 196

B. 197

C. 198

D. 199

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $a = \underbrace{111.....1}_{55 \text{ times}}, b = 1+10+10^2+10^3+10^4$ and $c = 1+10^5+10^{10}+....+10^{50}$, then

$b, \frac{a}{2}, c$ are in
A. arithmetic progression

$b, \sqrt{a}, c$ are in
B. geometric progression

a, b, c are in geometric
C. progression

$a, \sqrt{b}, c$ are in
D. arithmetic progression

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Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The term independent of x in the expansion of $\left[(t^{-1}-1)x+\left(t^{-1}+1\right)^{-1}x^{-1}\right]^{8}$ is :

A. $56\left(\frac{1-t}{1+t}\right)^{3}$

B. $56\left(\frac{1+t}{1-t}\right)^{3}$

C. $70\left(\frac{1-t}{1+t}\right)^{4}$

D. $70\left(\frac{1+t}{1-t}\right)^{4}$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For any real θ, the maximum value of $\cos^2(\cos\theta)+\sin^2(\sin\theta)$ is

A. 1

B. $1+\sin^2 1$

C. $1+\cos^2 1$

D. $1/2$

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Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The remainder when $(2021)^{2023}$ is divided by 7 is

A. 2

B. 3

C. 4

D. 5

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $a_1, a_2, a_3 \ldots\ldots\ldots\ldots$ be terms on A.P. If $\frac{a_1+a_2+\ldots\ldots\ldots a_p}{a_1+a_2+\ldots\ldots\ldots+a_q} = \frac{p^2}{q^2}, p \neq q$, then $\frac{a_6}{a_{21}}$ equals

A. $\frac{41}{11}$

B. $\frac{7}{2}$

C. $\frac{2}{7}$

D. $\frac{11}{41}$

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\tan\theta = 3\tan\phi$, then the maximum value of $\tan^2(\theta - \phi)$ is (where, $\tan\phi > 0$)

A. 1

B. $\frac{1}{3}$

C. 2

D. 4

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P. then $9^{ax+1}, 9^{bx+1}, 9^{cx+1}, x \neq 0$ are in:

A. G.P.

B. G.P. only if $x < 0$

C. G.P. only if $x > 0$

D. none of these

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Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the expansion of $\left(x^{\frac{4}{5}}+x^{-\frac{1}{5}}\right)^{n}$, the coeficient of the 8th and 19th terms are equal. The term independent of x is given by

A. C_{21}^{27}

B. C_{20}^{25}

C. C_{21}^{25}

D. C_{22}^{27}

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $\frac{3}{4}+\frac{15}{16}+\frac{63}{64}+\ldots$ upto n terms is

A. $n-\frac{4^{n}}{3}-\frac{1}{3}$

B. $\mathrm{n}+\frac{4^{-\mathrm{n}}}{3}-\frac{1}{3}$

C. $n+\frac{4^{n}}{3}-\frac{1}{3}$

D. $\mathrm{n}-\frac{4^{-\mathrm{n}}}{3}+\frac{1}{3}$

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Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = x^3 + ax^2 + bx + c$ be polynomial with integer coefficients. If the roots of $f(x)$ are integer and are in Arithmetic Progression, then 'a' cannot take the value

A. -642

B. 1214

C. 1323

D. 1626

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the sum of odd numbered terms and the sum of even numbered terms in the expansion of $(x + a)^n$ are A and B respectively, then the value of $(x^2 - a^2)^n$ is

A. $A^2 - B^2$

B. $A^2 + B^2$

C. $4AB$

D. None

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Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The coefficient of x^{20} in
$(1+x)(1+2x)(1+4x)(1+8x)\ldots\left(1+2^{20}x\right)$ is

A. $2^{211}-2^{190}$

B. $2^{191}-2^{171}$

C. $2^{231}-2^{209}$

D. $2^{161}-2^{142}$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $2\sin^2\theta+4\cos(\theta+\alpha)\sin\alpha\sin\theta+\cos2\ (\alpha+\ \theta)$ is

A. $\cos\theta+\cos\alpha$

B. independent of θ

C. independent of α

D. independent of both θ and α

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Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The coefficient of x^{50} in the polynomials after parenthesis have been removed and like terms have been collected in the expansion
$(1+x)^{1000}+x(1+x)^{999}+x^2 \quad (1+x)^{998}+\ldots+x^{1000}$ is $\frac{\lambda!}{\mu!v!}$, then the value of $\lambda+2\mu+3v$ must be $(v>\mu)$

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

For $n \in N$, in the expansion of $\left(\sqrt[4]{x^{-3}}+a\sqrt[4]{x^5}\right)^n$, the sum of all the binomial coefficients lie between 200 and 400. Also, the term independent of x is 448, then the value of a is

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

If a, b, c be last three digits of the number $(6!+1)^{6!}$, then the value of $a+b+c$ is

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Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

The numerical value of $\sin\frac{\pi}{18}\cdot\sin\frac{5\pi}{18}\cdot\sin\frac{7\pi}{18}$ is equal to k, find value of 16k:

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

If $(1+3+5+\ldots.+p)+(1+3+5+\ldots+q)=(1+3+5+\ldots$ r) where each set contains the sum of consecutive odd integers then the smallest possible value of $p+q+r$ is

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